Freight Rate Prediction — Solution Report

ML Engineer Assessment · HGBR log(rate-per-mile) · seed=42

1. Summary

Predict posted_rate (USD) for 12,000 loads (Nov–Dec 2025) using 48,000 labeled loads (Jan–Oct 2025).
Model: HistGradientBoostingRegressor on log(rate-per-mile); prediction = exp(output) × distance.
CV: 3 rolling-origin folds. Best model avg non-outlier WAPE = 3.34% vs 10.74% naive baseline (−68%).

2. Data and Quality Issues

train-test.csv: 48,000 rows (Jan–Oct 2025). validation.csv: 12,000 rows (Nov–Dec 2025, no posted_rate).

Issue	Scale (train / val)	Treatment
Weight negative	292 / 145	abs()
Weight missing	300 / 165	Median by equipment
Weight right-censored	1,191 / —	Kept (informative ceiling)
market_index missing	374 / 249	Per-date median, inputs only
quote_signal missing	0 / 0	No action
RPM outliers (<1.2/>4.5)	670 / —	Excluded training; kept in eval
Duplicate load_ids	0 / 0	No action
New cities in val	8 cities	OOV '__unknown__'; lat/lon intact
Road dist vs haversine	Mean +170 mi	Both features; diff = detour proxy

3. Train / Validation Split

Why random K-fold is wrong: holdout is strictly future (Nov–Dec 2025). K-fold would mix future labels into training folds, inflating CV estimates. Strategy: rolling-origin expanding window (3 folds).

Fold	Training window	Validation window	n train	n val
1	Jan–Jun 2025	Jul–Aug 2025	28,806	9,671
2	Jan–Jul 2025	Aug–Sep 2025	33,718	9,429
3	Jan–Aug 2025	Sep–Oct 2025	38,477	9,523

4. Model

Target — log(RPM): RPM = posted_rate/distance is right-skewed. log(RPM) stabilises variance; prediction = exp(output) × distance (always positive).

Algorithm — HistGradientBoostingRegressor (sklearn): handles mixed features natively.
Params: max_iter=500, lr=0.05, max_depth=6, min_samples_leaf=20, l2_reg=0.1, seed=42.

What else was compared:
(a) Naive: median RPM × distance → avg non-outlier WAPE 8.76%
(b) HGBR squared_error [SELECTED] → avg non-outlier WAPE 3.34%
(c) HGBR absolute_error → avg non-outlier WAPE 3.38%

Features (22): distance, haversine_dist, road−hav gap, weight_clean, market_index_filled, quote_signal, quote×dist, quote×hav, day_of_week, day_of_month, is_weekend, week_of_month, pickup_lat/lon, delivery_lat/lon, lat_diff, lon_diff, equipment_enc, pickup_enc, delivery_enc, lane_enc. month_of_year and day_of_year excluded — trees cannot extrapolate to unseen months.

5. Results

3-fold rolling-origin CV averages. 'All rows' includes 670 outliers. Non-outlier: RPM in [1.20, 4.50]/mi.

Model	Rows	MAE ($)	RMSE ($)	MAPE (%)	MedAPE (%)	WAPE (%)
Naive baseline	All rows	255.75	677.70	11.42	8.02	10.74
Naive baseline	Non-outlier	206.04	306.26	9.22	7.89	8.76
HGBR/sq_err ✓	All rows	130.13	630.37	5.81	2.62	5.47
HGBR/sq_err ✓	Non-outlier	78.49	132.27	3.57	2.58	3.34
HGBR/abs_err	All rows	131.05	630.80	5.97	2.46	5.50
HGBR/abs_err	Non-outlier	79.38	134.07	3.73	2.42	3.38

Per-fold detail — HGBR/squared_error, non-outlier rows:

Fold	Val window	n val	MAE	RMSE	MAPE	MedAPE	WAPE
1	Jul–Aug 2025	9,538	76.70	117.09	3.84	2.08	3.27
2	Aug–Sep 2025	9,298	74.97	134.04	3.49	2.47	3.20
3	Sep–Oct 2025	9,381	83.80	145.68	3.36	3.20	3.55
Avg	—	—	78.49	132.27	3.57	2.58	3.34

6. Fixed December Chart

Assumption: market index and quote signal filled with per-date median from validation.csv inputs for Dec 2025.
Lexington: (36.99152, −84.99876); Fort Wayne: (41.31561, −85.36206) from train-test.csv. Range: $785.49–$803.38 (mean $795.59).

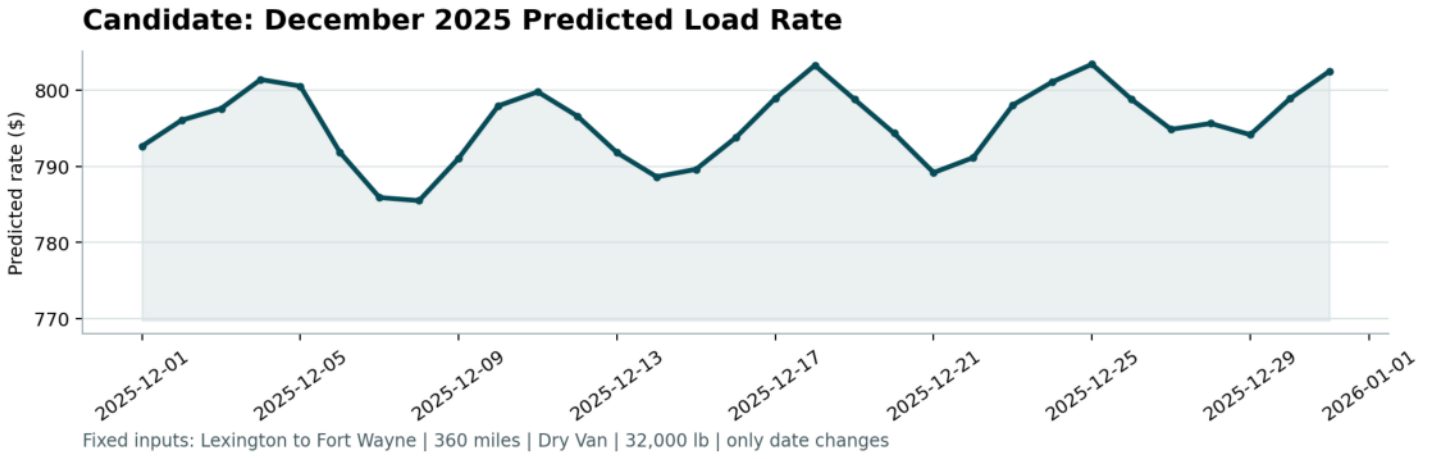


Figure 1: Dec 2025 predicted rate — Lexington→Fort Wayne, 360 mi, Dry Van, 32,000 lb. Generated by score.py.

7. Reproducibility

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All commands run from ML_assignment/ repo root:

  # 1. Install
  pip install -r requirements.txt

  # 2. Train, cross-validate, generate all prediction files
  python src/train.py

  # 3. Validate and generate December chart
  python score.py \
      --predictions validation_predictions.csv \
      --december-predictions december_chart_inputs_completed.csv \
      --output-dir scorer_results

  # 4. Regenerate report
  python make_report_docx.py

seed=42 in every HGBR call. Two independent runs produced byte-for-bit identical outputs
(max difference across all files = 0.0).
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